Causal Inference with Ordinal Outcomes: Two Approaches Using Density Estimation Techniques

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Abstract

Ordinal outcomes, such as Likert-type survey responses, are ubiquitous in political science, yet they pose distinctive challenges for causal inference. These outcomes are routinely analyzed by assigning numerical scores and applying linear models, despite longstanding concerns that ordinal responses do not identify treatment effects on an underlying cardinal scale without additional assumptions. Building on the classical identification result for ordinal outcomes and existing recommendations to report probability-based quantities, I develop a flexible likelihood-based framework for estimating distributional causal effects, including category-specific and cumulative treatment effects, which are invariant to admissible transformations of the latent scale. Existing approaches typically estimate these quantities using parametric ordered logit or ordered probit models, which rely on restrictive assumptions about the latent error distribution and index specification. This paper proposes a flexible likelihood-based estimation framework for ordinal causal inference using conditional normalizing flows. The proposed estimator flexibly learns the conditional distribution underlying observed ordinal responses without imposing a parametric latent error distribution, while preserving the likelihood-based structure familiar from classical ordered response models. The same framework accommodates both structured latent-index models and fully nonparametric conditional distribution estimators, allowing researchers to balance interpretability and modeling flexibility within a unified estimation procedure. Monte Carlo simulations show that the proposed estimator maintains low bias across a broad range of latent error distributions and nonlinear response mechanisms under which conventional ordered response models perform poorly. Replications of two published survey experiments demonstrate that flexible distributional estimation can materially alter substantive conclusions about experimental treatment effects.

1. Introduction

Causal inference has become a central organizing principle in political science. Experimental and quasi-experimental designs, including survey experiments, difference-in-differences, and regression discontinuity designs, are now standard empirical strategies for studying the effects of policies, institutions, and information. Survey experiments are particularly prominent because random assignment provides a transparent basis for causal identification. Between 2021 and 2024, roughly 20% of articles published in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* employed a survey experiment as part of their empirical strategy.

Many of these studies investigate outcomes that are inherently latent, including support for redistribution (Alt and Iversen, 2017; Magni, 2021), evaluations of political leaders (Canes-Wrone and De Marchi, 2002; Kriner and Schwartz, 2009), and foreign policy preferences (Tomz and Weeks, 2020). Although these concepts are commonly viewed as lying on an underlying continuous attitude scale, they are almost always observed through ordinal response categories such as Likert-type scales ranging from *strongly disagree* to *strongly agree*. Consequently, causal inference in political science routinely relies on outcomes that preserve the ordering of responses but not the distances between them.

Applied researchers typically analyze ordinal outcomes in one of three ways. The most common approach assigns numerical scores to response categories and estimates treatment effects using difference-in-means or ordinary least squares, implicitly treating ordinal responses as cardinal. Another common strategy collapses multiple categories into a binary outcome before estimation, sacrificing information about intermediate responses. A third approach fits ordered response models, such as ordered logit or ordered probit, which explicitly account for the ordinal nature of the data through a latent-variable representation. Each approach reflects a different set of assumptions about the relationship between the observed ordinal responses and the underlying latent attitude.

This paper begins from a classical observation in the ordinal response literature: latent treatment effects are identified only up to location and scale transformations of the underlying latent variable. Consequently, treatment effects defined through arbitrary numerical codings of ordinal responses generally require additional measurement assumptions linking the observed categories to a cardinal latent scale (Schröder and Yitzhaki, 2017; Bond and Lang, 2018; Bloem, 2022). This observation motivates a focus on distributional causal quantities, such as category-specific and cumulative treatment effects, which are defined directly on the observed ordinal responses and do not rely on assumptions about the cardinal spacing between adjacent categories.

This perspective complements recent recommendations in political methodology to summarize ordinal treatment effects using predicted probabilities rather than latent coefficients (Hanmer and Ozan Kalkan, 2013). Rather than revisiting how ordered response models should be interpreted, this paper asks how these probability-based causal quantities should be estimated when the assumptions underlying conventional ordered response models are difficult to justify. The methodological focus therefore shifts from interpreting latent coefficients to flexibly estimating the conditional ordinal response distribution. Existing semiparametric estimators relax some of these assumptions but remain difficult to apply with rich covariate structures or nonlinear response mechanisms that are common in modern survey research.

Estimating these distributional causal effects requires modeling the conditional distribution of ordinal responses. Ordered logit and ordered probit remain the workhorses of applied political science because they efficiently exploit the ordering of response categories, admit interpretable latent-variable representations, and often perform well when their assumptions are approximately satisfied. Their validity, however, depends on assumptions about the latent error distribution and index specification that may be difficult to justify in many survey experiments. Political attitudes are frequently polarized or clustered, producing multimodal latent distributions, while persuasive treatments often affect moderate respondents differently from strong supporters, generating nonlinear treatment-response relationships. In such settings, misspecification can distort estimated category probabilities and cumulative treatment effects, even when the experimental design itself identifies the causal effect. The practical challenge is therefore not causal identification, but flexible estimation of the conditional ordinal response distribution.

This paper develops a flexible likelihood-based framework for estimating causal effects with ordinal outcomes. The proposed estimator employs conditional normalizing flows to model the conditional response distribution without imposing a fixed parametric latent error distribution. Unlike existing semiparametric estimators, the framework remains computationally feasible with high-dimensional covariates while preserving exact likelihood-based inference. A single estimation procedure spans classical latent-index models, generalized latent-index specifications, and fully nonparametric conditional response distributions, allowing researchers to balance structural assumptions and modeling flexibility while targeting the same probability-based causal quantities.

I evaluate the proposed framework through Monte Carlo simulations designed to reflect empirically realistic survey settings, including polarized latent attitudes, nonlinear treatment-response relationships, and non-Gaussian latent error distributions. Across these settings, the proposed estimator maintains low bias while conventional ordered response models exhibit substantial distortions in estimated probability effects. I then revisit the survey experiments of Tomz and Weeks (2020) and Mattingly et al. (2025), showing that flexible estimation of the ordinal response distribution can materially alter substantive conclusions regarding the magnitude and distribution of treatment effects.

The remainder of the paper proceeds as follows. Section 2 formalizes causal inference with ordinal outcomes and motivates distributional causal estimands. Section 3 reviews existing estimation approaches and introduces the proposed likelihood-based framework. Section 4 evaluates finite-sample performance through Monte Carlo simulations. Section 5 presents the empirical applications. Section 7 concludes.

2. Causal Inference with Ordinal Outcomes

This section develops the conceptual framework underlying the remainder of the paper. I begin by describing ordinal survey responses as measurements of an underlying latent attitude using the classical threshold-crossing model. I then revisit a well-known identification result showing that latent treatment effects are identified only up to location and scale, and discuss its implications for causal inference with ordinal outcomes. Finally, I introduce distributional causal estimands that are directly identified from observed ordinal responses and motivate the estimation framework developed in the next section.

2.1. Latent Outcomes, Ordinal Responses, and Causal Effects

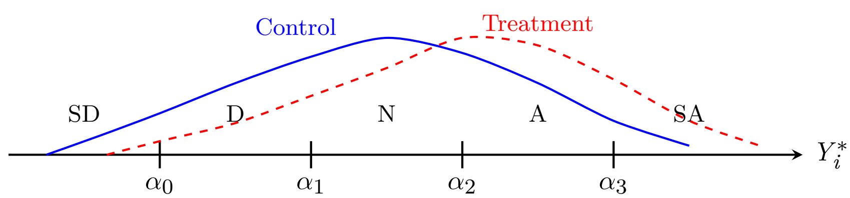


FIGURE 1. Ordinal responses arise by partitioning an underlying latent attitude into ordered response categories. The observed data identify how treatment changes probabilities of falling into response categories, but not the absolute scale of the latent attitude.

Many quantities of interest in political science, such as support for redistribution (Alt and Iversen, 2017; Magni, 2021), immigration preferences (Mayda and Rodrik, 2005), trust in institutions, and foreign policy attitudes (Scheve and Slaughter, 2001; Tomz and Weeks, 2020), are naturally viewed as continuous latent constructs. Survey respondents, however, rarely report these attitudes on a continuous scale. Instead, researchers observe ordinal responses, such as Likert items ranging from *strongly disagree* to *strongly agree*, ordered approval categories, or frequency scales. Consequently, empirical analyses are conducted using outcomes that preserve the ordering of respondents' attitudes but not the distances between adjacent response categories.

Following the classical ordered response framework, let Y_i^* denote respondent i's latent attitude and let $Y_i \in 0, 1, \ldots, J$ denote the observed ordinal response. The observed response is generated through a threshold-crossing mechanism,

$$Y_i = j \quad \Longleftrightarrow \quad \alpha_{j-1} < Y_i^* \leq \alpha_j, \qquad j = 0, \ldots, J,$$

where

$$-\infty = \alpha_{-1} < \alpha_0 < \cdots < \alpha_J = \infty$$

are unknown thresholds partitioning the latent scale. Figure 1 illustrates this measurement process. Throughout the paper I assume respondents share a common reporting function. Section XX discusses extensions allowing reporting thresholds to vary across respondents following generalized ordered response models (Williams, 2006). Alternative survey designs, such as anchored vignettes (King et al., 2004; King and Wand, 2007), provide complementary approaches for addressing heterogeneous reporting behavior.

The threshold-crossing representation should be interpreted as a measurement model rather than a structural model. It formalizes the idea that ordinal responses preserve the ordering of the underlying attitudes while discarding information about the distances between adjacent categories. Consequently, the observed responses provide only partial information about the latent construct: they reveal which interval of the latent scale a respondent occupies, but not the respondent's exact location within that interval.

To describe causal effects, let $Y_i^*(d)$ denote the latent potential outcome under treatment level d, and let

$$Y_i(d) = g\left(Y_i^*(d)\right)$$

denote the corresponding observed ordinal potential outcome. Throughout the paper I adopt the standard assumptions of SUTVA, conditional ignorability, and positivity (Rubin, 1974). If the latent outcome Y_i^* were directly observed, these assumptions would suffice to identify conventional causal estimands such as

$$ATE^* = \mathbb{E} = \left[Y_i^*(1) - Y_i^*(0)\right].$$

In practice, however, researchers observe only the ordinal response Y_i. The central question is therefore not whether causal effects are identified, but which causal quantities remain identified once the underlying latent attitude is observed only through ordered response categories. The next subsection revisits the classical identification result for ordinal response models and uses it to distinguish between causal quantities that require additional measurement assumptions and those that do not.

2.2. Identification of Causal Effects from Ordinal Outcomes

The threshold-crossing framework provides a useful representation of ordinal outcomes, but it also reveals a fundamental limitation of what can be learned from ordinal data. A classical result in the ordered response literature is that latent-variable models are identified only up to location and scale. Although this normalization is standard in econometrics and psychometrics, its implications for causal inference have received comparatively little attention in applied political science. In particular, survey experiments frequently interpret treatment effects estimated from numerically coded ordinal responses as effects on an underlying latent attitude without making explicit the measurement assumptions required for such an interpretation.

To formalize this point, suppose the latent outcome satisfies

$$Y_i^* = h_\theta(X_i) + D_i^\top \tau + \varepsilon_i, \tag{1}$$

where $h(X_i, \beta)$ denotes the systematic component of the latent outcome, τ is the treatment effect on the latent scale, and ε_i has cumulative distribution function F. Combining this latent-index specification with the threshold-crossing measurement model implies

$$\begin{aligned} P(Y_i = j \mid X_i, D_i) &= F\left(\alpha_j - h(X_i, \beta) - D_i^\top \tau\right) \\ &= F\left(\alpha_{j-1} - h(X_i, \beta) - D_i^\top \tau\right). \end{aligned}$$

Now consider any constants $a \in \mathbb{R}$ and $c > 0$, and define a transformed latent outcome

$$\tilde{Y}_i^* = a + cY_i^*,$$

together with transformed thresholds

$$\tilde{\alpha}_j = a + c\alpha_j.$$

Because both the latent outcome and the thresholds are transformed simultaneously, the probabilities of the observed ordinal responses remain unchanged.

Theorem 1 (Location and Scale Nonidentification). *Under the threshold-crossing latent-variable model, any positive affine transformation of the latent outcome and thresholds generates the same distribution of the observed ordinal responses. Consequently, treatment effects on the latent outcome are identified only up to location and scale.*

Theorem 1 should not be interpreted as a failure of causal identification. Under the standard assumptions of SUTVA, conditional ignorability, and positivity, causal effects remain identified for any well-defined outcome. Rather, the theorem highlights a measurement problem. Ordinal responses preserve the ordering of latent attitudes but do not reveal the cardinal scale on which those attitudes are measured. Consequently, causal quantities defined on the latent scale require additional measurement assumptions beyond those needed for causal identification itself.

This distinction has immediate implications for empirical practice. Different analyses of ordinal outcomes target different causal quantities and therefore rely on different measurement assumptions. For example, assigning numerical scores to ordinal categories and estimating mean differences implicitly assumes that those scores provide a meaningful cardinal representation of the latent attitude. By contrast, causal quantities defined directly on the observed ordinal responses do not require assumptions about the cardinal spacing between adjacent categories. The next subsection formalizes these probability-based causal estimands and shows how they arise naturally from the observed ordinal response distribution.

2.3. Distributional Causal Estimands

The identification result in Theorem 1 does not imply that causal inference with ordinal outcomes is fundamentally limited. Rather, it suggests that inference should focus on causal quantities defined directly on the observed ordinal responses. Such quantities require no assumptions about the cardinal spacing of the latent attitude and therefore remain invariant to admissible transformations of the latent scale. This perspective is closely aligned with recommendations to report predicted probabilities rather than latent coefficients when interpreting ordinal response models (Hanmer and Ozan Kalkan, 2013). Here I formalize these quantities as causal estimands and show that they are identified under the standard assumptions of causal inference.

The fundamental object is the distribution of potential ordinal outcomes. Let

$$\pi_j(d) = P\left(Y_i(d) = j\right), \qquad j = 0, \ldots, J,$$

denote the probability that a randomly selected individual reports category j under treatment level d.

Under the standard assumptions of SUTVA, conditional ignorability, and positivity, these probabilities are identified by standardization,

$$\pi_j(d) = \mathbb{E}_X\left[P\left(Y_i = j \mid D_i = d, X_i\right)\right],$$

where the expectation is taken over the covariate distribution of the target population. In randomized experiments, this expression simplifies to the observed proportion of respondents falling into category j within each treatment arm.

Proposition 1. *Under SUTVA, conditional ignorability, and positivity, the distribution of potential ordinal outcomes is identified by standardization. Consequently, any causal quantity that is a functional of this distribution is also identified.*

This proposition has an important practical implication. Once the conditional ordinal response distribution has been estimated, a wide range of causal quantities become available without fitting additional models. The primary statistical task is therefore estimation of the conditional response probabilities,

$$P(Y_i = j \mid D_i, X_i),$$

which serve as the building blocks for all distributional causal estimands considered in this paper.

The simplest causal quantity is the category-specific treatment effect,

$$\Delta_j = \pi_j(1) - \pi_j(0),$$

which measures how treatment changes the probability that a respondent selects category j. For example, in a survey experiment on support for military intervention, $\Delta_{\text{Strongly Agree}}$ measures the change in the proportion of respondents expressing the strongest level of support.

In many political science applications, however, interest centers on whether respondents exceed a substantively meaningful threshold rather than on a single response category. This motivates cumulative treatment effects,

$$\Delta_{\geq j} = P\left(Y_i(1) \geq j\right) = P\left(Y_i(0) \geq j\right),$$

which quantify changes in the probability of responding at or above category j. For example, a researcher may wish to estimate whether an informational treatment increases the probability that respondents express at least moderate support for a policy. Unlike dichotomizing the outcome before estimation, cumulative treatment effects can be calculated for every threshold after estimating the ordinal response distribution, allowing the full information contained in the ordinal responses to be retained.

These estimands also clarify the relationship between the proposed framework and existing empirical practice. Assigning numerical scores to ordinal categories estimates

$$\mathbb{E} = \left[s(Y_i(1)) - s(Y_i(0))\right],$$

where $s(\cdot)$ denotes the chosen scoring rule. This quantity is well defined for a fixed coding rule but generally does not identify a treatment effect on the underlying latent attitude. Similarly, dichotomizing an ordinal outcome estimates one cumulative treatment effect corresponding to a single threshold while discarding information about the remaining response categories. Estimating the full ordinal response distribution instead makes both category-specific and cumulative treatment effects available simultaneously, allowing researchers to select the quantity most appropriate for their substantive question after estimation rather than before it.

The discussion above highlights an important feature of distributional causal inference. Rather than targeting a single summary measure, the primary object of interest is the conditional distribution of ordinal responses,

$$P(Y_i = j \mid D_i, X_i),$$

from which a wide range of causal quantities can be obtained. Category-specific treatment effects and cumulative treatment effects are the primary estimands considered in this paper because they are directly interpretable and commonly reported in applied political science. More generally, however, any functional of the estimated response distributions may be calculated after estimation, including global measures of distributional change such as the one-dimensional Wasserstein (earth mover's) distance. Consequently, the statistical problem addressed in the remainder of this paper is estimation of the conditional ordinal response distribution rather than any single causal summary.

3. Flexible Estimation of Ordinal Response Distributions

Section 2 established that the identified causal quantities of interest are functionals of the conditional ordinal response distribution,

$$P(Y_i = j \mid D_i, X_i),$$

including category-specific and cumulative treatment effects. The remaining statistical problem is therefore estimation of this conditional distribution. This section first reviews the assumptions underlying existing estimation approaches and then develops a flexible likelihood-based framework that relaxes these assumptions while retaining the interpretability and probabilistic structure of classical ordered response models.

3.1. Existing Estimation Approaches

The dominant approach to estimating ordinal response distributions in political science is the ordered response model, most commonly ordered logit or ordered probit. These models combine a latent-index representation with a threshold-crossing measurement model. Specifically, the latent outcome is assumed to satisfy Equation (1), and the observed ordinal response is generated through the threshold-crossing mechanism introduced in Section 2.1. The distinction between ordered logit and ordered probit lies in the assumed distribution of the latent error term, with the former assuming a standard logistic distribution and the latter assuming a standard normal distribution.

These models remain the workhorses of applied political science for good reason. They naturally respect the ordinal nature of survey responses, provide interpretable latent-variable representations, admit likelihood-based inference, and often perform well when their assumptions are approximately satisfied. Moreover, predicted probabilities and first differences derived from ordered response models have become the preferred quantities for interpreting ordinal treatment effects in political methodology (Hanmer and Ozan Kalkan, 2013).

The validity of these probability estimates, however, depends on the assumptions used to construct the underlying response distribution. First, ordered response models assume a fixed parametric distribution for the latent error term. Second, they typically impose a single-index structure in which all covariate effects operate through the same latent index. Together, these assumptions imply a particular shape for the conditional ordinal response distribution.

In many survey experiments these assumptions may be difficult to justify. Political attitudes are frequently polarized or clustered, producing multimodal latent distributions. Ceiling and floor effects can compress observed responses near the boundaries of Likert scales. Persuasive treatments often affect moderate respondents differently from strong supporters, generating nonlinear treatment-response relationships. Although ordered response models may continue to provide useful approximations in many applications, misspecification of the latent error distribution or index structure can distort the estimated category probabilities and cumulative treatment effects that constitute the primary quantities of substantive interest.

A number of extensions have been proposed to relax these assumptions. Generalized ordered response models allow covariate effects to vary across response thresholds, while semiparametric estimators relax parametric assumptions on the latent error distribution. Bayesian nonparametric approaches provide additional flexibility through hierarchical modeling. Existing approaches, however, typically relax only one component of the model or become computationally difficult in the presence of high-dimensional covariates. The next subsection introduces a unified likelihood-based framework that accommodates both structured latent-index models and fully nonparametric conditional response distributions within a common estimation procedure.

3.2. Why Conditional Normalizing Flows?

The previous subsection showed that estimation of the conditional ordinal response distribution is the central statistical problem underlying causal inference with ordinal outcomes. The question, therefore, is not simply how to relax the parametric assumptions of ordered logit and ordered probit, but how to do so while retaining the probabilistic structure that makes these models attractive. An ideal estimator should remain likelihood-based, accommodate high-dimensional covariates, recover the full conditional response distribution, and allow researchers to incorporate as much or as little structural modeling as their application warrants.

Existing semiparametric approaches relax some of the assumptions imposed by classical ordered response models, most commonly the distribution of the latent error term. Kernel-based estimators and related semiparametric methods provide important robustness gains, but they typically become computationally demanding as the dimension of the conditioning variables increases. Modern survey experiments routinely adjust for rich collections of demographic characteristics, pretreatment covariates, and treatment interactions, making high-dimensional conditional density estimation an increasingly important practical concern.

Conditional normalizing flows provide a natural framework for addressing these challenges. A normalizing flow represents an unknown probability distribution through a sequence of invertible transformations of a simple base distribution. Unlike discriminative machine-learning methods that directly predict response probabilities, normalizing flows estimate the underlying conditional density through maximum likelihood. Consequently, the estimated model remains fully probabilistic: likelihood-based estimation, model comparison, and probabilistic calibration are all retained within a flexible distributional framework.

A second advantage is that the same framework accommodates different levels of structural modeling. At one extreme, the latent outcome may follow the familiar single-index specification used in classical ordered response models, with the normalizing flow used only to estimate a flexible latent error distribution. At the other extreme, the systematic component of the latent outcome may itself be modeled nonparametrically, allowing the entire conditional response distribution to be learned with minimal functional-form restrictions. Rather than forcing researchers to choose between a fully parametric model and a fully nonparametric estimator, the proposed framework provides a continuum of models that differ only in the amount of structure imposed on the conditional response distribution.

Finally, because the proposed estimator targets the conditional response distribution itself, the resulting model is not tied to any particular causal summary. Once the distribution has been estimated, category-specific treatment effects, cumulative treatment effects, and other functionals of the ordinal response distribution can all be obtained as post-estimation quantities without fitting additional models. In this sense, the conditional ordinal response distribution serves as the fundamental statistical object, while the causal quantities discussed in Section 2.3 arise as different summaries of that common object.

3.3. A Flexible Likelihood-Based Estimation Framework

The previous subsection argued that the central statistical task is estimation of the conditional ordinal response distribution. I retain the latent-variable representation in Equation (1), but depart from classical ordered response models in how its components are specified. Rather than fixing both the systematic component and the latent error distribution, the proposed framework allows each to be modeled flexibly while preserving a likelihood-based estimation procedure.

The framework is built around two components: the systematic component of the latent outcome, represented by the function $h(\cdot)$, and the latent error distribution F. Existing estimators correspond to particular choices of these components. For example, ordered probit specifies a linear index together with a standard normal error distribution, while ordered logit replaces the normal distribution with a standard logistic distribution. Semiparametric ordered response models relax the distributional assumption on the latent error while retaining a structured index specification.

The proposed framework generalizes these approaches by allowing flexibility in either or both components of the latent outcome model. When substantive theory suggests that treatment and covariates act through a common latent index, the function $h(\cdot)$ may be specified parametrically while the latent error distribution is estimated flexibly. Alternatively, when the functional relationship between the covariates and the latent outcome is itself uncertain, the systematic component may also be estimated nonparametrically. In this way, the same estimation framework spans classical latent-index models, semiparametric ordered response models, and fully flexible conditional distribution estimators.

The key ingredient is a conditional normalizing flow, which provides a flexible parameterization of unknown probability distributions while retaining exact likelihood evaluation. Rather than assuming that the latent error follows a predetermined distribution, such as the normal or logistic, the proposed estimator learns this distribution from the data through a sequence of invertible transformations. Because these transformations admit tractable likelihood evaluation, estimation proceeds by maximum likelihood in much the same way as conventional ordered response models. A particularly attractive feature of ordinal response models is that the latent error is scalar. When combined with monotone spline flows, this permits exact evaluation of both the latent density and cumulative distribution function, so that the resulting estimator retains the likelihood structure of classical ordered response models without numerical approximation.

This likelihood-based construction provides several practical advantages. First, it directly targets the conditional ordinal response distribution, from which all causal quantities introduced in Section 2.3 can be recovered as post-estimation summaries. Second, unlike kernel-based semiparametric estimators, the framework remains computationally feasible in the presence of high-dimensional covariates that are common in modern survey experiments. Finally, because the same likelihood encompasses both structured and fully flexible specifications, researchers can vary the amount of structural modeling while preserving a common inferential framework.

The following subsections describe the conditional normalizing-flow construction and the corresponding likelihood in detail.

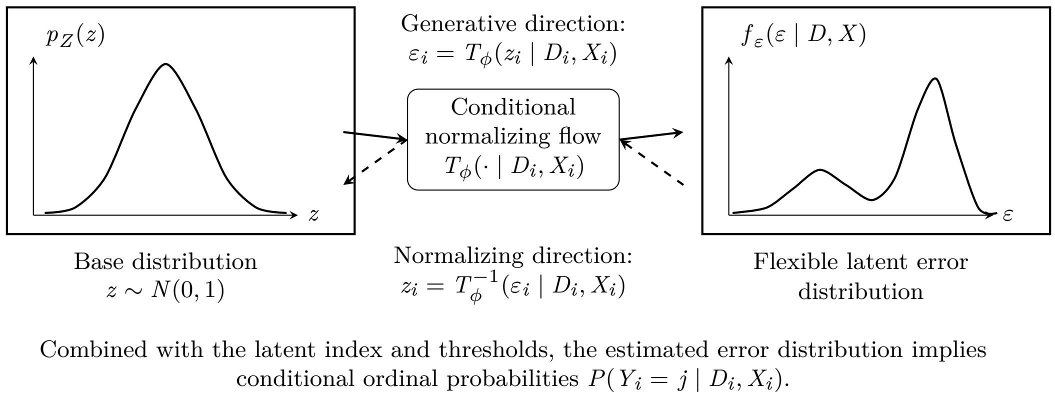


FIGURE 2. A conditional normalizing flow transforms a simple base distribution into a flexible latent error distribution. In the proposed ordered-response framework, this replaces the fixed normal or logistic error assumption while retaining likelihood-based estimation.

3.4. Modeling the Latent Error Distribution with Conditional Normalizing Flows

The proposed framework replaces the fixed parametric latent error distribution with a flexible conditional density estimator. Rather than assuming that the latent error follows a standard normal or logistic distribution, I estimate its conditional distribution directly from the data while retaining exact likelihood-based inference.

Normalizing flows provide a natural framework for this task. Originally introduced by **?**, normalizing flows represent an unknown probability distribution through a sequence of invertible transformations of a simple base distribution. Because each transformation is invertible, probability densities remain analytically tractable through the change-of-variables formula, allowing estimation to proceed by maximum likelihood. Since their introduction, normalizing flows have become a standard approach for flexible likelihood-based density estimation (**??**).

Figure 2 illustrates the basic idea. Let

$$z_i \sim p_Z(z)$$

denote a simple base distribution, taken throughout this paper to be the standard normal distribution. A conditional normalizing flow transforms this base distribution into the latent error distribution through an invertible mapping,

$$\varepsilon_i = T_\phi(z_i \mid D_i, X_i),$$

where the transformation is parameterized by ϕ and conditioned on the treatment and covariates. The same transformation may be inverted,

$$z_i = T_\phi^{-1}(\varepsilon_i \mid D_i, X_i),$$

allowing observations to be mapped back to the base distribution. This invertibility is the defining feature of normalizing flows and enables exact likelihood evaluation through the standard change-of-variables formula,

$$f_\varepsilon(\varepsilon_i \mid D_i, X_i) = p_Z(z_i) \left| \det \frac{\partial T_\phi^{-1}(\varepsilon_i \mid D_i, X_i)}{\partial \varepsilon_i} \right|.$$

Unlike kernel density estimators, the transformation is represented by a parametric function that scales naturally to high-dimensional conditioning variables. Throughout this paper I construct $T_\phi(\cdot)$ using monotonic rational-quadratic spline (RQS) transformations (Durkan et al., 2019). RQS flows provide smooth monotone mappings while retaining closed-form evaluation of both the inverse transformation and the Jacobian (**?**). Because the latent error is one-dimensional, this construction also permits exact evaluation of the implied cumulative distribution function, which is required to compute the threshold probabilities in ordered response models.

The conditional flow is used only to model the latent error distribution. The systematic component of the latent outcome in Equation (1) remains independent of this choice and may therefore be specified at different levels of structural complexity. In the structured specification, the function $h(\cdot)$ is parameterized using a conventional latent-index model, so that flexibility is introduced only through the latent error distribution. This specification may be viewed as a semiparametric generalization of ordered logit and ordered probit. In the fully flexible specification, both the systematic component and the latent error distribution are modeled nonparametrically, allowing the conditional ordinal response distribution to be learned with minimal functional-form assumptions. Both specifications share the same likelihood and estimation procedure, differing only in the amount of structure imposed on the latent outcome model.

3.5. Likelihood-Based Estimation

The latent-index representation in Equation (1), together with the threshold-crossing measurement model introduced in Section 2.1, implies that the probability of observing response category j is

$$\begin{aligned} P(Y_i = j \mid D_i, X_i) &= P\left(\alpha_{j-1} < Y_i^* \le \alpha_j \mid D_i, X_i\right) \\ &= F_\phi\left(\alpha_j - h(X_i) - D_i^\top \tau \mid D_i, X_i\right) - F_\phi\left(\alpha_{j-1} - h(X_i) - D_i^\top \tau \mid D_i, X_i\right), \end{aligned}$$

where $F_\phi(\cdot)$ denotes the conditional cumulative distribution function implied by the normalizing flow introduced in the previous subsection. This expression differs from ordered logit and ordered probit only through the choice of F_ϕ. Rather than assuming a logistic or normal distribution, the latent error distribution is estimated directly from the data.

Also, Unlike general multivariate flow models, the one-dimensional monotone spline construction used here admits exact evaluation of the conditional cumulative distribution function F_ϕ. Consequently, the category probabilities required by the ordered-response likelihood can be computed analytically without numerical integration.

Assuming independent observations, the log-likelihood for the observed ordinal responses is

$$\ell(\Theta) = \sum_{i=1}^{n} \sum_{j=0}^{J} 1(Y_i = j) \log P(Y_i = j \mid D_i, X_i),$$

where

$$\Theta = \tau, h, \phi, \alpha$$

collects all unknown parameters of the latent outcome model, the conditional normalizing flow, and the reporting thresholds.

The parameters are estimated by maximizing this likelihood using stochastic gradient optimization. Because the conditional normalizing flow admits exact evaluation of both the latent density and its cumulative distribution through the invertible transformation, the resulting objective remains a genuine likelihood rather than a surrogate prediction loss. Consequently, estimation retains many of the desirable features of conventional ordered response models, including likelihood-based model comparison and probabilistic calibration, while allowing substantially greater flexibility in the latent error distribution.

The likelihood in Equation (1) encompasses both structured and flexible specifications discussed in the previous subsection. When the systematic component $h(\cdot)$ is parameterized linearly, the estimator may be viewed as a semiparametric generalization of ordered logit and ordered probit. When $h(\cdot)$ is itself modeled nonparametrically, the same likelihood estimates the conditional ordinal response distribution under minimal functional-form assumptions. In either case, the fitted model directly yields the conditional probabilities

$$P(Y_i = j \mid D_i, X_i),$$

which serve as the basis for estimating all causal quantities introduced in Section 2.3.

3.6. Estimation and Recovery of Causal Quantities

The proposed estimator is obtained by maximizing the likelihood in Section 3.5 with respect to the parameters governing the systematic component, the reporting thresholds, and the conditional normalizing flow. Because the objective is an exact likelihood, estimation proceeds using standard stochastic gradient optimization and automatic differentiation. Throughout the paper, optimization is performed using the Adam algorithm with minibatch stochastic gradient descent until convergence of the log-likelihood.[1]

The fitted model yields estimates of the conditional ordinal response distribution,

$$\widehat{P}(Y_i = j \mid D_i, X_i),$$

for every response category, treatment level, and covariate value. These estimated probabilities form the fundamental output of the procedure. Unlike conventional ordered response models, the proposed estimator is not designed to recover a single coefficient or treatment effect. Instead, it estimates the conditional response distribution itself, from which a variety of causal quantities may be obtained.

For example, the estimated category-specific treatment effect is obtained by standardization,

$$\widehat{\Delta}_j = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{P}(Y_i = j \mid D_i = 1, X_i) - \widehat{P}(Y_i = j \mid D_i = 0, X_i)\right],$$

which corresponds to the plug-in estimator of the g-formula (Robins, 1986). Cumulative treatment effects are computed analogously,

$$\widehat{\Delta}_{\geq j} = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{P}(Y_i \geq j \mid D_i = 1, X_i) - \widehat{P}(Y_i \geq j \mid D_i = 0, X_i)\right].$$

More generally, because the entire conditional response distribution is available after estimation, any functional of that distribution may be computed without fitting additional models. Examples include average predicted probabilities, first differences, stochastic dominance measures, or global summaries of distributional change such as the one-dimensional Wasserstein (earth mover's) distance. The proposed estimator therefore separates estimation from interpretation: the model is fitted once, while substantive quantities of interest are obtained as post-estimation summaries of the estimated response distribution.

The simulation study and empirical applications that follow focus primarily on category-specific and cumulative treatment effects because these quantities are directly interpretable and commonly reported in applied political science. The same fitted model, however, supports a substantially broader class of distribu-

[1] Implementation details, including network architecture, learning-rate schedules, initialization, and convergence diagnostics, are provided in Appendix XX.

tional summaries whenever alternative scientific questions arise. Throughout the simulations and empirical applications, uncertainty is quantified using the nonparametric bootstrap, which naturally propagates estimation uncertainty from the fitted conditional response distribution to all post-estimation functionals.

4. Monte Carlo Simulation

This section evaluates the finite-sample performance of the proposed estimation framework. The objective is not to demonstrate that flexible estimation universally outperforms classical ordered response models. Ordered logit and ordered probit remain efficient estimators when their assumptions are approximately satisfied. Rather, the simulations investigate the conditions under which the additional flexibility of the proposed estimator becomes important for recovering the conditional ordinal response distribution and the corresponding distributional causal effects.

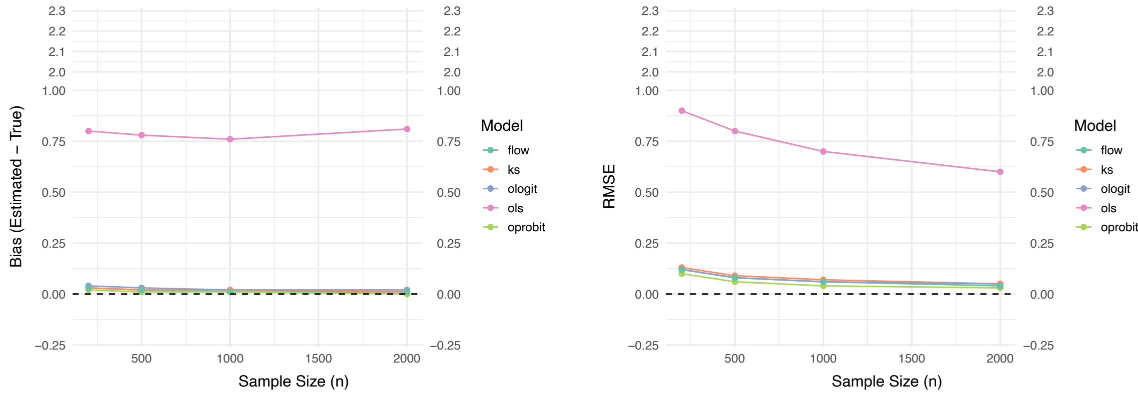
The simulation study considers a sequence of data-generating mechanisms designed to reflect empirical features commonly encountered in political science applications, including heavy-tailed and multimodal latent attitudes, nonlinear treatment-response relationships, and high-dimensional covariate structures. Across these settings, I compare the proposed estimator with conventional ordered response models and a fully nonparametric empirical estimator based on observed category frequencies. Performance is evaluated in terms of recovery of the conditional response distribution and the resulting causal quantities introduced in Section 2.3.

4.1. Simulation Design

All simulations are generated from the latent-index representation introduced in Equation (1), followed by the threshold-crossing measurement model described in Section 2.1. Unless otherwise stated, treatment assignment is completely randomized, ensuring that the identification assumptions developed in Section 2 hold by construction. Consequently, differences in performance arise solely from the estimation procedure rather than from failures of causal identification.

The simulation design systematically varies the two components of the latent outcome model that distinguish the proposed framework from conventional ordered response models: the systematic component $h(\cdot)$ and the latent error distribution F. The baseline scenario follows the assumptions of ordered probit, with a linear latent index and normally distributed errors. Additional scenarios progressively relax these assumptions by introducing heavy-tailed or multimodal latent error distributions, nonlinear latent-index functions, and higher-dimensional covariate structures representative of modern survey experiments. This design allows the simulations to isolate the consequences of each modeling assumption while remaining closely connected to the motivating examples discussed in the introduction.

For each data-generating process, I generate $R = 100$ Monte Carlo replications with sample size n. Each simulated dataset is analyzed using four estimators: ordered probit, ordered logit, an empirical estimator based on observed category frequencies, and the proposed conditional normalizing-flow estimator. The empir-



(a) (b)

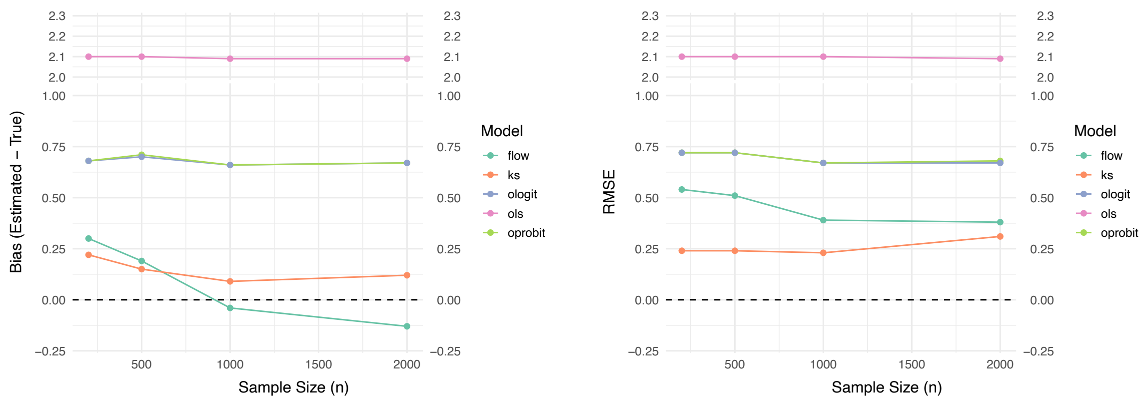
FIGURE 3. Monte Carlo Simulation Results: Standard Normal Error

ical estimator provides a nonparametric benchmark in randomized experiments by estimating treatment-arm response distributions directly from the observed data without imposing any model structure. Comparing these four estimators therefore illustrates the trade-off between modeling assumptions, statistical efficiency, and robustness to misspecification.

Simulation performance is evaluated from two perspectives. First, I examine how accurately each method recovers the conditional ordinal response distribution. Second, I evaluate the resulting category-specific and cumulative treatment effects obtained by regression standardization. Together, these metrics distinguish errors arising from poor distributional estimation from those arising in the post-estimation causal summaries that are ultimately reported by applied researchers.

Figure 1 summarizes the Monte Carlo simulation results for the standard normal error term case. Panel (a) shows the mean bias from 100 simulations of OLS, ordered logit model, ordered probit model, KDE-based estimator, and Normalizing flow based estimator. The biases of the KDE-based and NF-based estimators are comparable to the true model, the ordered probit. Panel (b) shows RMSE for each estimator, and again both KDE-based estimator and NF-based estimator perform comparable to ordered probit model. Considering that the standard normal error case meets the distributional assumption for ordered probit model, the great performance of the ordered probit model is expected. Also, the CDF of standard logistic is not too different from that of standard normal, ordered logit model also retains small bias. This demonstrates strong finite sample performance of both KDE-based and NF-based estimators.

Figure 2 describes the results from the lognormal-distribution error term case. Again OLS shows the worst results, clearly overestimating the treatment effect. Ordered probit and ordered logit resulted in very similar estimates and almost indistinguishable in the figure. However, these standard ordinal regression models also show large over estimation of NLTE, and the bias dies not shrink as the sample size increases. This clearly



(a) (b)

FIGURE 4. Monte Carlo Simulation Results: Lognormal distribution Error

demonstrates that they can be inconsistent estimators if the distributional assumptions are violated. Both of the flexible ordinal estimators introduced in this paper shows much better performance in terms of bias, and as expected, the biases seem to converge to 0 as the sample size increases. One panel (b) we observe a similar story, while OLS has very RMSE, standard ordered logit and probit maintains the RMSE around 0.7 to 0.74. Both KDE-based and NF-based estimators achieves the best performance in terms of RMSE.

Overall, the Monte Carlo simulation demonstrates that cardinalization approach depends on arbitrary labels and thus unreliable to capture the causal effect, standard ordered regression models may perform poorly when the distributional assumptions are violated. KDE-based estimator and NF-base estimator, however, demonstrate better performance across two cases and different sample sizes.

5. Application 1: Tomz and Weeks (2020)

The first example replicates the analysis by Tomz and Weeks (2020). The study looked into the question of how the human rights practices of foreign adversaries affect domestic public support for war through survey experiments. The main argument was that in democracies with well-established human rights, people are less willing to go to war with other countries that also respect human rights than with comparable countries that violate them. Thus they are focusing on two factors that can impact people's support for war: the political system of the foreign country and how much the foreign country respect human rights.

The paper uses total of 5 surveys, but this paper focus on the first, which contains an experiment for their main result. The survey was conducted by YouGov with a nationally representative sample of 1,430 US adults in October 2012. The treatment was given as in a form of vignettes on a country that is developing nuclear weapons. The vignettes also include trade and military relationship of the country and the US, and its conventional military strength which is set to the half of that of the US. In the vignettes, information about the political system (democracy = 1, not = 0) and human rights practices (respect = 1, not = 0) were binary treatments and they were randomized independently.

The outcome of interest here is the support for physical strike of the US armed forces to the country. Specifically, respondents were asked to answer in 5-point scale: *Favor strongly, Favor somewhat, Neither favor nor oppose, Oppose somewhat*, and *Oppose strongly.* In the paper, the authors take the cardinalization approach and OLS. They first recode the outcome to take numeric values from 1 to 5, then they used this to run OLS. As discussed in Section 2, the cardinalization is arbitrary and we only can interpret the ATE as suggested by authors when the cardinalization is capturing the unknown transformation function, which is untestable.

The model they used in their main results (model 2 of their Table 1) can be expressed as following:

$$Y^* = \tau_{DEM} \cdot DEM + \tau_{HR} \cdot HR + X^T \beta + \varepsilon$$

$$Y \in \{1, 2, 3, 4, 5\}$$

where Y^* is the support for the military strike in latent scale and Y is the observed outcome with cardinalization from 1 to 5, DEM and HR are binary variables indicating political system treatment and human rights practices treatment respectively, X for the control variables and ε for the error term.

Since they are interested in whether citizens living in democracy where human rights are respected (the US, to be more specific) are more reluctant to support military strike to a country who is developing nuclear weapons if the country also respecting human rights or the country has democratic political system, the causal parameters we are interested in the above equation are τ_{HR} and τ_{DEM}. We only can observe the ordinal outcome, we can only identify the effects up to scale, and here I adopted error-variance normalization we discussed in the previous section. I replicate their result on Table S3 of their paper, and use ordered logit, ordered probit and KDE-based estimator as a comparison.

Table 1 reports the results across all five specifications. Substantively, all models agree on the direction of the treatment effects: information that a country respects human rights or is a democracy significantly decreases domestic support for military action. However, the interpretation and statistical information utilized differ. The original OLS interpretation (a 12.6 percentage point drop for the Human Rights treatment) relies on the validity of the binarization and the cardinal 0–100 scale. By collapsing the 5-point scale into two categories, the original analysis discards potentially valuable information regarding the intensity of respondent preferences.

	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
Respect HR	-12.6^{***}	-0.46^{***}	-0.49^{***}	-0.48	-0.49^{***}
	(1.47)	(0.10)	(0.06)	(0.27)	(0.06)
Democracy	-9.5^{***}	-0.31^{**}	-0.34^{***}	-0.33	-0.34^{***}
	(1.47)	(0.10)	(0.06)	(0.22)	(0.06)

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

TABLE 1. Replication of Tomz and Weeks (2020) Table S3. ATE represents the original binarized OLS estimates (0-100 scale). NLTE columns report standardized latent coefficients ($\sigma = 1$).

In contrast, NLTE models utilize the full ordinal range of the data. The standardized coefficients remain remarkably consistent across the Ordered Probit, KDE-based, and NF-based models. For instance, the effect of respecting human rights is estimated at approximately -0.49 standard deviations across these specifications. The fact that the flexible semiparametric estimators (KDE and NF) closely track the Ordered Probit results serves as a diagnostic, suggesting that the latent error distribution for this specific survey approximates a normal distribution. In this instance, the parametric assumption appears robust, though the semiparametric approach provides the necessary validation that the original cardinalization was not distorting the underlying direction of the effects.

6. Application 2: Mattingly et al. (2025)

The second application replicates the large-scale cross-national study by Mattingly et al. (2025), which examines how authoritarian regimes utilize state-produced media to promote their political and economic models to foreign audiences. We focus our replication on the authors' primary hypotheses: (H1) US government messaging increases preference for the American model, (H2) Chinese Communist Party (CCP) messaging increases preference for the Chinese model, and (H3) in the presence of competing messages, public preference shifts toward the Chinese model.

To test the hypothesis, authors launched multiple survey experiments took place in 19 different countries (N = 6,000) in June 2022. Respondents were assigned via block randomization to one of four conditions: viewing US state-produced videos (USA), CCP-produced videos (China), a combination of both (Competition), or nature-themed placebo videos (Control). The primary outcomes were measured using a 6-point Likert scale, ranging from *Strongly Prefer the US* (1) to *Strongly Prefer China* (6). The original analysis relied on OLS regression, which cardinalizes these ordinal categories by treating the 1–6 scale as an interval measure.

After the treatment, respondents were asked to answer two questions which measures main outcome variables (whether they prefer the US (political / economic) system or (political / economic) Chinese system) in 6 categories from *Strongly Prefer the US* to *Strongly Prefer China*. The cardinalize the outcomes by assigning numeric values, so that *Strongly Prefer the US* was recoded as 1 and *Strongly Prefer China* was recoded as 6. The original results came from OLS results with these cardinalized outcomes.

The model Mattingly et al. (2025) used for their main results can be expressed as follows

$$Y_i^* = \tau_{CHINA} \cdot CHINA_i + \tau_{USA} \cdot USA_i + \tau_{COMP} \cdot COMP_i + X_i^T \gamma + \varepsilon_i$$

$$Y \in \{1, 2, 3, 4, 5, 6\}$$

where $CHINA_i$, USA_i and $COMP_i$ are indicator variables for the respective experimental conditions, and X_i is a vector of pre-treatment covariates including age, gender, education and family income.

While the original study estimates the Average Treatment Effect (ATE) by applying OLS directly to the categorical labels (1–6), our approach focuses on identifying the latent treatment parameters (τ). As is standard in semiparametric ordinal models, the latent scale is identified only up to location and scale; hence, we apply the error-variance normalization ($\sigma = 1$). This allows for a direct comparison between the parametric specifications (Ordered Probit and Logit) and the flexible KDE and NF-based estimators. Under this framework, the coefficients represent the shift in the latent preference distribution in units of standard deviations, providing a measure of treatment intensity that is robust to the arbitrary cardinalization of the 6-point scale.

Table 2 reports the replication of the original OLS results, and results of NLTE estimation using different ordinal regression models including KDE-based model and flow-based model. For preferences over the political system (upper panel), we find a stark divergence between parametric and semiparametric models. While the direction of the effects remains consistent across all specifications, the magnitude of the estimates is significantly inflated in the cardinal (OLS) and parametric ordinal (Logit/Probit) models. Specifically, the estimated effect of the Chinese treatment in the NF-based model (0.37) is nearly two-thirds smaller than the OLS estimate (1.04) and less than half the size of the Ordered Probit estimate (0.76).

Notably, the *Competition* effect, which the original study identified as a robust driver of preference for the Chinese model (0.36, $p < 0.001$), is substantially attenuated in the flexible models. In the NF-based specification, this effect drops to 0.11, suggesting that when the assumption of a Normal error distribution is relaxed, the persuasive power of competitive rhetoric appears much more fragile than previously reported.

This suggests that the OLS and Probit models may be over-attributing shifts in the latent distribution to the treatment, rather than accounting for the non-normal dispersion of political preferences.

In contrast, the lower panel of Table 2 reports preferences over economic systems, where the models show greater consensus. While OLS still yields the largest point estimates, the gap between the Probit (0.57) and the NF-based model (0.58) for the China treatment is negligible. This indicates that the latent distribution of economic preferences in this sample likely approximates normality more closely than political preferences. However, the OLS estimates for all treatments remain consistently higher than their latent counterparts, reinforcing the argument that cardinalizing Likert scales may lead to substantially different conclusions.

Outcome: Preference over Political System					
	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
China	1.04***	0.73***	0.76***	0.36***	0.37***
	(0.05)	(0.04)	(0.04)	(0.07)	(0.04)
USA	−0.43***	−0.35***	−0.38***	−0.18***	−0.17***
	(0.04)	(0.04)	(0.04)	(0.05)	(0.03)
Competition	0.36***	0.21***	0.25***	0.13*	0.11**
	(0.05)	(0.04)	(0.04)	(0.06)	(0.04)
Outcome: Preference over Economic System					
	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
China	0.87***	0.56***	0.57***	0.59***	0.58***
	(0.05)	(0.04)	(0.04)	(0.07)	(0.06)
USA	−0.57***	−0.39***	−0.43***	−0.42***	−0.43***
	(0.05)	(0.04)	(0.04)	(0.05)	(0.05)
Competition	0.28***	0.16***	0.18***	0.19**	0.18**
	(0.06)	(0.04)	(0.04)	(0.06)	(0.06)

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

TABLE 2. Replication of Mattingly et al. (2025) Table C3. ATE represents original OLS estimates. NLTE columns report standardized latent coefficients from various ordinal specifications.

7. Conclusion

This paper has examined the problem of causal inference when the outcome of interest is measured on an ordinal scale, a situation that arises frequently in political science research on attitudes and preferences. Building on a standard latent-variable framework, we showed that, even under conventional causal assumptions such as SUTVA, ignorability, and positivity, treatment effects on the underlying continuous outcome are in general identified only up to scale. The joint distribution of ordinal responses is invariant to positive affine transformations of the latent outcome and thresholds, so the absolute magnitude and location of the latent average treatment effect cannot be recovered from the data alone. Any attempt to compare coefficients (across models or across studies) must therefore adopt an explicit normalization. To address this, we proposed the *Normalized Latent Treatment Effect (NLTE)* as an alternative causal estimand, ensuring that latent treatment effects are expressed in a transparent and comparable metric.

We argued that common strategies for analyzing ordinal outcomes are problematic in light of this limitation. Cardinalization, which assigns numeric scores to categories and applies tools such as OLS or difference-in-means, produces estimates whose size and sometimes even sign can depend sensitively on arbitrary labeling choices (Schröder and Yitzhaki, 2017; Bond and Lang, 2018; Bloem, 2022). Binarization discards information about intermediate categories, leading to inefficient estimators and potentially obscuring substantively important changes in the distribution of responses. Parametric ordinal models, including ordered logit and ordered probit, avoid arbitrary numeric labels and respect the ordered nature of the data, but rely on strong assumptions about the latent error distribution. When these assumptions are violated the resulting estimators converge to pseudo-true parameters and can suffer substantial bias (Manski, 1988; Greene and Hensher, 2010; Johnston, McDonald and Quist, 2020; Smits et al., 2020).

This paper introcused two estimators that retain the interpretability of latent treatment effects while relaxing parametric assumptions on the error distribution. The first is a semiparametric KDE-based estimator that uses kernel density estimation to approximate the latent error CDF and constructs a quasi-likelihood for the ordinal model. The second employs normalizing flows to model the error distribution as an invertible transformation of a simple base distribution within a maximum likelihood framework. Both estimators treat the latent outcome as parametric, estimate the error distribution flexibly from the data.

Monte Carlo simulations demonstrated that these choices matter in practice. When the latent error distribution is correctly specified (for example, normal for ordered probit), parametric ordinal models perform well, as expected. However, under misspecification such as lognormal error, ordered logit and probit can be noticeably biased, whereas the KDE-based and NF-based estimators remain consistent and exhibit favorable finite-sample performance. The simulations also showed that cardinalization yields highly unstable estimates across different labeling schemes, and that binarization is less efficient and can miss meaningful shifts among categories. In this sense, the density-estimation-based approaches provide a robust alternative that better respects the ordinal nature of the data and the limited information it contains about the latent scale.

The empirical applications further underscore the substantive importance of these methods. Reanalyzing Tomz and Weeks (2020) on human rights and military force, we found that standard binarized OLS tends to understate the magnitude of the latent treatment effect. Conversely, our replication of Mattingly et al. (2025) regarding authoritarian propaganda reveals that the perceived "superiority" of Chinese state cues over American cues in the original study is, in part, a methodological artifact. By accounting for the floor effects and non-normal dispersion of political preferences, our proposed estimators show that the relative persuasive power of these cues is much closer than cardinal OLS suggests. These examples demonstrate how the assumption of cardinality or normality can inadvertently shape—and potentially distort—substantive conclusions in top-tier political science research.

Several directions for future work follow from these results. First, while we focused on single-item ordinal outcomes, extending density-estimation-based methods to multi-item settings and integrating them more tightly with IRT-style measurement models would be valuable. Second, developing practical diagnostic tools for assessing error distribution misspecification in ordinal models, building on surrogate residuals or related techniques, would complement the estimators proposed here. Third, applications with clustered, panel, or hierarchical data structures pose additional challenges for both identification and estimation that merit further study. More broadly, the analysis reinforces the importance of taking the ordinal nature of many political science outcomes seriously and of being explicit about the assumptions and normalizations that underlie estimates of causal effects on latent attitudes.

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8. Appendix

A. KDE-Based Ordinal Regression

This appendix provides additional details for the semiparametric KDE-based estimator introduced in Section 3. We first define the quasi-likelihood, then describe the kernel estimation of the error distribution and state a consistency result.

A.1. Model and Quasi-Likelihood

Recall the latent outcome model

$$Y_i^* = f(X_i, \beta) + D_i^\top \tau + \varepsilon_i,$$

and the threshold-based reporting function

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \le \alpha_j, \qquad j = 0, 1, \ldots, J,$$

with $-\infty = \alpha_{-1} < \alpha_0 < \cdots < \alpha_J = \infty$. For notational simplicity, define the latent outcome

$$V_i(\beta, \tau) = f(X_i, \beta) + D_i^\top \tau.$$

Let F denote the (unknown) CDF of ε_i. Conditional on (X_i, D_i), the probability of observing category j is

$$p_j(X_i, D_i; \alpha, \beta, \tau, F) = F\big(\alpha_j - V_i(\beta, \tau)\big) - F\big(\alpha_{j-1} - V_i(\beta, \tau)\big).$$

If F were known, the population log-likelihood would be

$$\ell(\alpha, \beta, \tau; F) = \mathbb{E}\left[\sum_{j=0}^{J} 1_{Y_i=j} \log p_j(X_i, D_i; \alpha, \beta, \tau, F)\right].$$

In the sample of size n, the infeasible sample log-likelihood is

$$\ell_n(\alpha, \beta, \tau; F) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=0}^{J} 1_{Y_i=j} \log p_j(X_i, D_i; \alpha, \beta, \tau, F).$$

When F is unknown, we replace it by a nonparametric estimator $\hat{F}$ constructed via KDE, obtaining a *quasi*-log-likelihood

$$\hat{\ell}_n(\alpha, \beta, \tau) = \frac{1}{n} \sum_{i=1}^{n} \hat{m}(X_i) \sum_{j=0}^{J} 1_{Y_i = j} \log \hat{p}_j(X_i, D_i; \alpha, \beta, \tau),$$

where

$$\hat{p}_j(X_i, D_i; \alpha, \beta, \tau) = \hat{F}\big(\alpha_j - V_i(\beta, \tau)\big) - \hat{F}\big(\alpha j - 1 - V_i(\beta, \tau)\big),$$

and $\hat{m}(X_i)$ is a trimming function that downweights observations in extreme regions of the covariate space where KDE may be unstable. The KDE-based estimator $(\hat{\alpha}, \hat{\beta}, \hat{\tau})$ is defined as any maximizer of $\hat{\ell}_n(\alpha, \beta, \tau)$.

A.2. Kernel Estimation of the Error Distribution

To construct $\hat{F}$, we exploit the relationship between the distribution of the index V_i and the error term ε_i. Let

$$g_1(v \mid Y \leq j) = \text{density of } V \text{ given } Y \leq j, \qquad g_0(v \mid Y > j) = \text{density of } V \text{ given } Y > j,$$

and let

$$\pi_j = \mathbb{P}(Y_i \leq j), \qquad 1 - \pi_j = \mathbb{P}(Y_i > j).$$

Then by the Bayes' rule,

$$\mathbb{P}(Y_i \leq j \mid V_i = v) = \frac{\pi_j g_1(v \mid Y \leq j)}{\pi_j g_1(v \mid Y \leq j) + (1 - \pi_j) g_0(v \mid Y > j)}.$$

Since $Y_i^* = V_i + \varepsilon_i$, and $Y_i \leq j$ iff $Y_i^* \leq \alpha_j$, we have

$$\mathbb{P}(Y_i \leq j \mid V_i = v) = \mathbb{P}\big(\varepsilon_i \leq \alpha_j - v\big) = F\big(\alpha_j - v\big).$$

Thus, if we can estimate π_j, $g_1(\cdot)$, and $g_0(\cdot)$ from the data, we can construct an estimator $\hat{F}$ such that

$$\hat{F}(\alpha_j - v) \approx \widehat{\mathbb{P}}(Y_i \leq j \mid V_i = v).$$

The probabilities π_j and $1 - \pi_j$ can be estimated by sample proportions:

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} 1_{Y_i \leq j}, \qquad 1 - \hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} 1_{Y_i > j}.$$

The conditional densities g_1 and g_0 are estimated using KDE. For concreteness, suppose we use a univariate kernel $K(\cdot)$ (e.g. Gaussian) and bandwidths that may depend on j. Let $n_1(j) = \sum_{i=1}^{n} 1_{Y_i \leq j}$ and $n_0(j) =$

$\sum_{i=1}^{n} 1_{Y_i > j}$. Define the subsamples

$$V_i : Y_i \leq j, \qquad V_i : Y_i > j,$$

and estimate the conditional densities as

$$\hat{g}_1(v \mid Y \leq j) = \frac{1}{n_1(j)h_{1j}} \sum_{i: Y_i \leq j} K\left(\frac{v - V_i}{h_{1j}}\right),$$

$$\hat{g}_0(v \mid Y > j) = \frac{1}{n_0(j)h_{0j}} \sum_{i: Y_i > j} K\left(\frac{v - V_i}{h_{0j}}\right),$$

where h_{1j} and h_{0j} are bandwidths chosen according to standard rules (possibly location-adaptive). Then the estimated conditional probability is

$$\widehat{\mathbb{P}}(Y_i \leq j \mid V_i = v) = \frac{\hat{\pi}_j \hat{g}_1(v \mid Y \leq j)}{\hat{\pi}_j \hat{g}_1(v \mid Y \leq j) + (1 - \hat{\pi}_j)\hat{g}_0(v \mid Y > j)}.$$

We interpret this as an estimator of the CDF of ε_i evaluated at $\alpha_j - v$:

$$\hat{F}(\alpha_j - v) \equiv \widehat{\mathbb{P}}(Y_i \leq j \mid V_i = v).$$

In practice, the KDE is implemented with local smoothing and trimming to reduce boundary bias and instability in regions with sparse data. Following Abramson (1982) and Silverman (1986), one can use pilot density estimates to define location-specific bandwidths and damping functions that prevent the bandwidth from shrinking too quickly in low-density regions. We adopt these techniques, as in Klein and Sherman (2002), to ensure $\sqrt{n}$ consistency.

A.3. Consistency of the KDE-Based Estimator

This section sketches the main consistency result for the KDE-based estimator. Fuller proofs follow from standard arguments combined with the consistency of locally smoothed KDEs (Klein and Spady, 1993; Abramson, 1982; Silverman, 1986).

Assumption 1 (Smoothness of the Index Distribution). *The conditional density of the index $V_i(\beta, \tau)$ given (X_i, D_i) exists, is strictly positive on its support, and is continuously differentiable up to a sufficiently high order. The support of V_i is compact or can be truncated to a compact set via trimming without affecting the asymptotic behavior of the estimator.*

Assumption 2 (KDE Regularity Conditions). *The kernel $K(\cdot)$ is a bounded, symmetric density with compact support. Bandwidths h_{1j} and h_{0j} satisfy $h_{1j} \to 0$, $h_{0j} \to 0$, and $nh_{1j} \to \infty$, $nh_{0j} \to \infty$ as $n \to \infty$, with additional conditions ensuring the consistency of locally smoothed KDEs (e.g. rates controlling the pilot bandwidth and damping parameters).*

Under these conditions, the KDE estimators $\hat{g}_1$ and $\hat{g}_0$ converge uniformly to g_1 and g_0, and the estimated CDF $\hat{F}$ converges uniformly to the true F on compact subsets of the support. Consequently, the quasi-log-likelihood $\hat{\ell}_n(\alpha, \beta, \tau)$ converges uniformly to $\ell_n(\alpha, \beta, \tau; F)$, and the maximizer of $\hat{\ell}_n$ converges to the maximizer of the infeasible log-likelihood (up to the usual positive affine transformation).

Theorem 2 (Consistency of the KDE-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions 1–2 hold. Then, as $n \to \infty$,*

$$\left|(\hat{\alpha}, \hat{\beta}, \hat{\tau}) - (\alpha, \beta, \tau)\right| = o_p(1),$$

up to an arbitrary positive affine transformation $(a+c\alpha, c\beta, c\tau)$ with $c > 0$. In particular, the latent treatment effects τ are consistently estimated up to scale.

Because the ordinal model is only identified up to scale, we impose the error-variance normalization $\text{Var}(\varepsilon_i) = 1$ ex post by rescaling the coefficients using the estimated error variance $\hat{\sigma}_\varepsilon^2$ as described in Section 3. Under this normalization, the KDE-based estimator yields treatment effects that are directly comparable to those from ordered probit, ordered logit (after rescaling), and the normalizing-flow-based estimator.

B. Normalizing Flows

This appendix provides additional detail on the normalizing-flow-based estimator described in Section 3. We briefly review the change-of-variables formula, define the ordinal likelihood with a learned error distribution, and state a consistency result.

B.1. Change-of-Variables Formula

Let Z be a random variable with a known base density $f_Z(z)$ (for example, the standard normal density), and let $T_\theta : \mathbb{R} \to \mathbb{R}$ be an invertible, differentiable transformation parameterized by θ. Define

$$\varepsilon = T_\theta(Z).$$

Then, by the change-of-variables formula, the density of ε is

$$f_\theta(\varepsilon) = f_Z\big(T_\theta^{-1}(\varepsilon)\big)\left|\frac{d}{d\varepsilon}T_\theta^{-1}(\varepsilon)\right|.$$

The corresponding CDF F_θ is

$$F_\theta(u) = \int_{-\infty}^{u} f_\theta(t), dt.$$

We refer to the family $T_{\theta\theta\in\Theta}$ as a *normalizing flow*. By choosing a sufficiently flexible family (e.g. coupling layers, spline flows), we can approximate a wide class of univariate distributions with tractable densities and gradients.

B.2. Ordinal Likelihood with a Flow-Based Error Distribution

We embed the flow-based error distribution into the latent outcome model,

$$Y_i^* = f(X_i,\beta) + D_i^\top\tau + \varepsilon_i, \qquad \varepsilon_i = T_\theta(Z_i), \quad Z_i \sim \mathcal{N}(0,1),$$

with thresholds α defining the observed ordinal outcome:

$$Y_i = j \quad \Longleftrightarrow \quad \alpha_{j-1} < Y_i^* \le \alpha_j.$$

Let f_θ and F_θ denote the density and CDF of ε_i implied by the flow T_θ. Conditional on (X_i, D_i), the probability of category j is

$$\mathbb{P}(Y_i = j \mid X_i, D_i; \alpha,\beta,\tau,\theta) = F_\theta\big(\alpha_j - V_i(\beta,\tau)\big) - F_\theta\big(\alpha_{j-1} - V_i(\beta,\tau)\big),$$

where $V_i(\beta,\tau) = f(X_i,\beta) + D_i^\top\tau$. The sample log-likelihood is

$$\ell_n(\alpha,\beta,\tau,\theta) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=0}^{J} 1_{Y_i=j}\log\left[F_\theta\big(\alpha_j - V_i(\beta,\tau)\big) - F_\theta\big(\alpha_{j-1} - V_i(\beta,\tau)\big)\right].$$

We estimate $(\alpha,\beta,\tau,\theta)$ jointly by maximizing $\ell_n(\alpha,\beta,\tau,\theta)$ with respect to all parameters. Because T_θ is invertible and differentiable, both f_θ and F_θ are tractable, and gradients of the log-likelihood with respect to $(\alpha,\beta,\tau,\theta)$ can be computed efficiently using automatic differentiation.

B.3. Consistency of the Flow-Based Estimator

This section briefly states a consistency result for the Normalizing-Flow-Based estimator. Let $(\alpha_0, \beta_0, \tau_0, F_0)$ denote the true parameters and error distribution. Suppose that the true error distribution F_0 belongs to the closure of the flow family $F_\theta : \theta \in \Theta$, and that the latent outcome model and threshold structure are correctly specified.

ASSUMPTION 3 (Flow Approximation and Identification). *The flow family $F_\theta : \theta \in \Theta$ is rich enough that there exists $\theta_0 \in \Theta$ such that $F_{\theta_0} = F_0$ (or F_{θ_0} approximates F_0 arbitrarily well). The parameter vector $(\alpha, \beta, \tau, \theta)$ is identified up to a positive affine transformation $(a + c\alpha, c\beta, c\tau)$ with $c > 0$.*

ASSUMPTION 4 (Regularity Conditions). *The parameter space for $(\alpha, \beta, \tau, \theta)$ is compact or can be restricted to a compact subset; the log-likelihood function $\ell_n(\alpha, \beta, \tau, \theta)$ satisfies standard regularity conditions (continuity, differentiability, integrability); and the model is correctly specified in the sense that the true data-generating process is in the model class for some $(\alpha_0, \beta_0, \tau_0, \theta_0)$.*

Under these conditions, standard maximum likelihood theory implies that the flow-based estimator is consistent (up to scale) and asymptotically normal.

THEOREM 3 (Consistency of the Flow-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions 3–4 hold. Then, as $n \to \infty$,*

$$\left|(\hat{\alpha}, \hat{\beta}, \hat{\tau}, \hat{\theta}) - (\alpha_0, \beta_0, \tau_0, \theta_0)\right| = o_p(1),$$

up to an arbitrary positive affine transformation $(a + c\alpha_0, c\beta_0, c\tau_0)$ with $c > 0$. In particular, the latent treatment effects τ_0 are consistently estimated up to scale.

As with the KDE-based estimator, we impose the error-variance normalization $\mathrm{Var}(\varepsilon_i) = 1$ ex post by rescaling the coefficients using the estimated error variance. Given $\hat{\theta}$, we estimate

$$\hat{\sigma}\varepsilon^2 = \mathrm{Var}\hat{F}\theta(\varepsilon_i) \approx \frac{1}{M} \sum m = 1^M \left(T_{\hat{\theta}}(Z_m)\right)^2,$$

where $Z_m \sim \mathcal{N}(0, 1)$ are independent draws from the base distribution. We then report

$$\hat{\tau}^{\text{unit-var}} = \frac{\hat{\tau}}{\hat{\sigma}\varepsilon}, \qquad \hat{\beta}^{\text{unit-var}} = \frac{\hat{\beta}}{\hat{\sigma}\varepsilon},$$

so that all treatment effects are expressed on the same unit-error-variance latent scale as in the main text.